13. Due to $14\frac{2}{7}\%$ fall in the price, 10 more bananas are available at Tk. 420.
    (a) If $14\frac{2}{7}\%$ of a number is 10, then determine the number.
    (b) What is the price of one dozen of banana at present?
    (c) To make a profit of $33\frac{1}{3}\%$, what should be selling price of each dozen of banana.

2·4 Unitary Method

Let the price of 10 ballpen be Tk. 50. Then, we can say easily that the price of one ballpen is Tk. $\frac{50}{10}$ or Tk. 5. Now, the price of any number of ballpens, can be determined from that of one ballpen. Such as, the price of 8 ballpen is Tk. (5×8) = Tk. 40.

Therefore, in unitary method, we can determine the price, weight, quantity of the definite number of things from the price, weight, quantity of a single thing. Let us observe the following examples.

Example 13. If the price of 7 dozens of pencils is Tk. 1442, then what is the price of one dozen of pencil ?

Solution : The price of 7 dozens of pencil is Tk. 1442

$\therefore$ " " " 1 " " " " Tk. $\frac{1442^{206}}{7}$ or Tk. 206

$\therefore$ The price of one dozen of pencil is Tk. 206.

Let us observe that Tk. 1442 has been divided by 7 to find the price of 1 dozen of pencils.

Example 14. 10 persons can do a piece of work in 7 days. In how many days can 5 persons do the same work ?

Solution : 10 persons can do the piece of work in 9 days

$\therefore$ 1 " " " " " " " 9×10 days = 90 days.

In this case, the work done by one person needs 9 times time. That is, one person can do this work in 90 days. Now, if 5 persons do the work, then the time needed is less than that of one person. That is, 5 persons can do the same work in $\frac{90}{5}$ days or 18 days.

Here, the time of one person is divided by 5, then we get the time of 5 persons.

Example 15. In a student mess of 50 students there is food for 4 days. How many days will do for 20 students with the same quantity of food ?

Solution : For 50 students the food lasts 4 days

$\therefore$ „ 1 „ „ „ „ (50×4) days or 200 days

$\therefore$ „ 20 „ „ „ „ $\frac{50 \times 4}{20}$ days or 10 days

Here, we see that the food of 50 students is enough for 4 days. The same quantity of food is enough for 200 days for one student. Again, it is enough for 20 students for 10 days. Thus it is seen that a reduced number of persons need more days; again an increased number of persons need less days.

Example 16. 20 workers can dig a pond in 15 days. How many workers can dig the pond in 20 days.

Solution :

In 15 days the pond is dug by 20 persons

$\therefore$ „ 1 „ „ „ „ „ „ 20×15 „

$\therefore$ „ 20 „ „ „ „ „ „ $\frac{20 \times 15}{20}$ „ or 15 persons

The number of required workers is 15.

Example 17. Walking 10 hours daily, Shafique crosses 480 km in 12 days. In how many days can he cross 360 km walking 10 hours daily?

Solution : Walking 10 hours daily

Shafique crosses 480 km in 12 days

$\therefore$ „ „ 1 km. „ $\frac{12}{480}$ „

$\therefore$ „ „ 360 km. „ $\frac{12 \times 360}{480}$ „ or 9 days

$\therefore$ Required time 9 days.

Example 18. A and B can do a piece of work in 12 days and 20 days respectively. In how many days A and B together can do that piece of work ?

Solution : A in 12 days can do a piece of work

$\therefore$ A „ 1 day „ „ $\frac{1}{12}$ part of a piece works

Again , B in 20 days can do a piece of work

$\therefore$ B „ 1 day „ „ $\frac{1}{20}$ parts of a piece work

$\therefore$ A & B together in 1 day can do $\left(\frac{1}{12}+\frac{1}{20}\right)$ part of a piece of work

$= \frac{5+3}{60}$ part

$= \frac{8}{60}$ part

$= \frac{2}{15}$ part

$\therefore$ A and B can do $\frac{2}{15}$ part of a piece of work in 1 day

$\therefore$ „ „ „ „ 1 part(total) „ „ $1 \div \frac{2}{15}$ or $1 \times \frac{15}{2}$ days

$= \frac{15}{2}$ days or $7\frac{1}{2}$ days

Required time $7\frac{1}{2}$ days.

Example 19. A 5-member family has provision for 20 days, with 40 kgs of rice. How many days will do for the same family with 70 kg of rice ?

Solution : For a 5-member family

40 kgs of rice is enough for 20 days

1 kg „ „ „ $\frac{20}{40}$ „

$\therefore$ 70 kgs of „ „ „ $\frac{\cancel{20} \times \cancel{70}^{35}}{\cancel{40}_{2}}$ days or 35 days

Required time 35 days.

Example 20 : A contractor made an agreement to complete a 100 km road in 20 days. Recruiting 250 labourers, he completed 62.5% construction work of the road in 10 days.

(a) If the first quantity is 62.5% of the second quantity. What will be second quantity : first quantity= $\square$?

(b) If 100 labourers were recruited, how many kilometres of road would have been constructed in 15 days?

(c) Show that the work will be finished ahead of 4 days of the scheduled time.

Solution :

a. Here, $62.50\% = \frac{\cancel{625}^{\cancel{25}}}{\cancel{10000}_{\cancel{1000}_{\cancel{40}_{8}}}}$

$\frac{62.50\%}{100} = \frac{\cancel{6250}}{\cancel{10000}} = \frac{5}{8}$

It means, 1st quantity is $\frac{5}{8}$ portion of the 2nd quantity.
If the 1st quantity is 5, the 2nd quantity will be 8
2nd quantity : the 1st quantity = 8:5

b. Here, 62.50% of 100 km

$$= \frac{100 \times 62.50}{100} \text{ km}$$
$$= 62.50 \text{ km}$$

∴ 250 labourers in 10 days completed 62.50 km of road.

∴ 1 labourer in 10 days completes $\frac{62.50}{250}$ km of road

∴ 1 labourer in 1 day completed $\frac{62.50}{200 \times 10}$ km of road

∴ 100 labourers in 15 days complete $\frac{62.50 \times 100 \times 15}{250 \times 10}$ km of road

$$= \frac{93750}{2500} \text{ km}$$
$$= 37.50 \text{ km}$$

If 100 labourers were recruited, 37.50 km of road would have been constructed in 15 days.

(c) From 'b' we get, 62.50% of 100 km = 62.50 km
100 labourers in 10 days construct 62.50 km of road
Remains (100-62.50) km of road
= 37.50 km of road
Time remains (20-10) days or, 10 days

∴ 250 labourers construct 62.50 km of road in 10 days.

∴ 250 labourers construct 1 km of road in $\frac{10}{62.50}$ days.

∴ 250 labourers construct 37.5 km of road in $\frac{10 \times 37.50}{62\ 50}$ days.

$$= \frac{\cancel{3750}^{6}}{\cancel{625}} = 6$$

∴ The work will be finished on the scheduled day (10-6) or 4 days ahead of the schedule time. (shown)

Exercise 2·3

1. In the following chart, match the expressions of the left to that of the right.

(a) Ratio	(a) %
(b) Unit ratio	(b) a fraction
(c) Symbol of percentage	(c) 1 : 5
(d) Ratio in greater inequality	(d) 9 : 9
(e) Ratio in less inequality	(e) 7 : 3

2. What is ratio ?
   a. A fraction b. An integer c . An odd number d. A prime number
3. Which is the equivalent ratio of 2 : 5 of the followings ?
   a. 2 : 3 b. 4 : 9 c. 4 : 10 d. 5 : 2
4. Which of the following is the compound ratio of 3 : 4 and 4 : 5?
   a. 15 : 16 b. 12 : 20 c. 7 : 9 d. 12 : 16
5. If the ratio 3 : 20 is expressed into percentage. Which of the following is correct?
   a. 3% b. 20% c. 15% d. 17%
6. 1% of 200 centimetre is –
   (a) 2 metre (b) 1 metre (c) 2 centimetre (d) 1 centimetre
7. In the ratio 1:5
   (i) the antecedent is 1 (ii) the subsequent is5 (iii) the inverse ratio is 5:1 Which is the correct answer?
   (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii
8. Of 100 boys and girls, if 60% is girls
   (i) the number of girls = 60
   (ii) the number of boys = 40
   (iii) the ratio of boys and girls = 3:2.